

MinJun Chang

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Education

Georgia Institute of Technology, PhD in Civil Engineering	Aug 2025 - Current
• Robotics and Intelligent Construction Automation Laboratory	
Yonsei University, BS in Mechanical Engineering	Mar 2019 - Feb 2025

Research

RICAL Group, Georgia Institute of Technology – Atlanta, GA, USA	Aug 2025 – Current
Graduate Research Assistant, Supervisor: Dr. Yong K. Cho	
• Deep-Learning based real-time human motion recognition with localization with IMU sensors in construction	
• Whole body control of a humanoid for dexterous tasks in construction and manufacture	
Dynamic Robotic System Laboratory, Seoul National University – Seoul, KR	Jul 2024 – Dec 2024
Undergraduate Research Assistant, Supervisor: Dr. Jaeheung Park	
• Model-free reinforcement learning framework with a state estimator neural-network fo bipedal locomotion	
Machine Learning and Control System Laboratory, Yonsei University – Seoul, KR	Jul 2022 – Jan 2023
Undergraduate Research Assistant, Supervisor: Dr. Jongeun Choi	
• Developed an enhanced localization algorithm with control input delay compensation	

Publications

Real-Time Temporal Convolution based Worker Motion Recognition with Three IMU Sensor Combinations, 43rd International Symposium on Automation and Robotics in Construction, <i>MinJun Chang</i> , Sai Machiraju, Francis Baek, Yong K. Cho†	Jun 2026
Field Evaluation of a Wearable Framework for Worker Motion Recognition and Zone-Level Localization, 43rd International Symposium on Automation and Robotics in Construction, <i>MinJun Chang</i> , S.K. Machiraju, Yong K. Cho†	Jun 2026
Large Language Models as Virtual Personas to Estimate Construction Practitioners' Perceptions of Robotic Automation, International Conference on Computing in Civil Engineering, Francis Baek, <i>MinJun Chang</i> , Eunbee Kim	Jun 2026
A Perception-to-Decision Vehicle Monitoring System for Highway Work-Zone Safety, International Conference on Computing in Civil Engineering, S.Y. Kim, <i>MinJun Chang</i> , S.K. Machiraju, Yong K. Cho†	Jun 2026
Robust Symmetric Bipedal Locomotion Development via Simultaneous State Estimator Neural Network Training, The 20th Korea Robotics Society Annual Conference, Poster <i>MinJun Chang</i> , Jaeyong Shin, Jaeheung Park†	Feb 2025
State prediction-based control input delay compensation for autonomous driving systems, The 18th Korea Robotics Society Annual Conference, Oral <i>MinJun Chang</i> , H.W. Nam, S.Y. Choi, J.H. Yang, J.H. Yang, Jongeun Choi†	Feb 2023

Patents and Copyrights

The Urine Examination Apparatus and Controlling Method of the Same KR10-2944428-00-00, Registered-Active	Mar 2026
Autonomous Driving Auxillary Cart Robot for Manufacture KR10-2024-0177135, under prosecution	Aug 2024
Eye Tuner: Media Literacy Program based on Pupil Tracking by Computer Vision Korea Copyright Commission, C-2024-039138, Registered-Active	Nov 2021

Professional Experience

Hanwha Systems(Space&Defense) , Satellite Software Engineer	Jan 2025 – May 2025
• ARMv7 MCU internal communication software programming for Small Synthetic Aperture Radar (SAR) Satellite	
GOLE Robotics , Robotics Engineer	Apr 2024 – Jun 2024
• Implemented global and local robot path planning algorithm on ROS2 for construction delivery robot WERO • Developed actuator controller package with C++/Python binding enabling python API usage of C++ source	
DRIMAES , Embedded Software Engineer	Oct 2022 – Mar 2024
• Linux, ARM MCU software/firmware programming for various products • Developed various communication protocols (Serial, MQTT, REST, CAN) • Implemented multiple virtual container management technique on cross-platform systems	
SIOT Infotech , Embedded Software Engineer	Feb 2022 – Oct 2024
• Developed embedded software for Mediatek, ESP chipsets and enhanced custom OpenWRT OS kernel	

Selected Awards

1st place, National ICT Smart Device Competition , Korean Ministry of Science and ICT	Aug 2024
• Awarded by the Minister of Science and ICT of Republic of Korea • Led a team of 5 in developing an Autonomous Manufacture Assistant CARTRASCHE	
1st place, 2022 Autonomous Driving Robot Racing Contest , Korean Robotics Society (KRoS)	Nov 2022
• Participated as Localization team member • Developed a control algorithm utilizing LiDAR, IMU, and GPS sensor fusion for collision avoidance	
Selection, Hanium Contest , Federation of Korea Information Industries	Nov 2021
• Led a team of 4 in developing Personalized Content Literacy program EYE-TUNER • Implemented pupil tracking algorithm for the program	
2nd place, Medical Hack 2021 , Busan City	Nov 2021
• Awarded by the mayor of Busan City • Implemented posture prediction algorithm with multiple load-cell sensors	
2nd place, Yonsei IHEI Workstation , Yonsei University	Jul 2020

Projects

Awaresite : Intelligent PPU System for Construction Productivity and Safety	Aug 2025 - Current
• Deep-Learning based worker motion recognition using IMU data • Zone scale indoor localization using BLE beacons	
CARTRASCHE: Autonomous Driving Auxiliary Cart Robot	Mar 2023 - Jul 2024
• Developed autonomous driving mobile robot with rotating shelf system using SLAM for navigation in ROS • Implemented custom RC filter and encoder-less motor control algorithm for activation • Managed the project flow and system overall management • Tools Used: C++, Python, ROS, LinuxOS	
UAV Fleet Management System for Robot Cluster in Factory	Aug 2023 - Dec 2023
• Implemented task scheduling and allocation algorithm based on order status for multi-robot network • Tools Used: Python, ROS, MQTT	
FennecBot: Industrial Anomaly Detection Mobile Robot	May 2023 - Aug 2023
• Developed multi-modal deep learning network for pipeline anomaly detection and the classification of pipeline leakage using RGB camera, and ultrasonic/acoustic sound camera • Operated on Scout mini with line-tracing algorithm detecting pipe leakage within Hyundai HI. factory	

Honors

Federico Stubbe Graduate Student Fellowship, Georgia Institute of Technology Aug 2025
Fellowship granted to competent graduate students

Next Generation Engineer, Institute for Promotion of Engineering and Science of Korea Dec 2024
Honor society for outstanding young engineers in Republic of Korea

Technologies

Programming: Python, C++ , C, JavaScript, MATLAB

Frameworks/Tools: ROS, IsaacSim/Lab, PyTorch, Docker, AWS, FastAPI, Solidworks, Fusion360

Hardware: ARM V7, Jetson Xavier, Jetson Nano, RaspberryPi, Arduino, STM32, ESP32, Unitree Go1

Languages: Korean(Native), English(Fluent), Japanese(Median)